

Data Communication

Content 07: Bandwidth Utilization – Multiplexing

Abdullah Ibne Masud Mahi (AIMM)

Department of Computer Science and Engineering
United International University



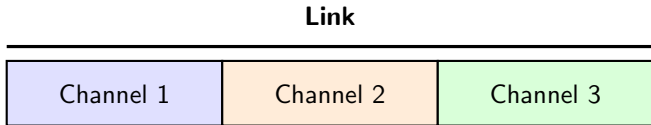
Why Multiplexing?

- Whenever the bandwidth of a medium linking two devices is **greater** than the bandwidth needs of the devices, the link can be **shared**.
- Building a separate physical link for every pair of communicating devices is expensive and wasteful.
- **Multiplexing (MUX)** is the set of techniques that allows the **simultaneous** transmission of multiple signals across a **single** data link.
- As data and telecommunication use increases, so does traffic – multiplexing lets us use existing bandwidth more efficiently.

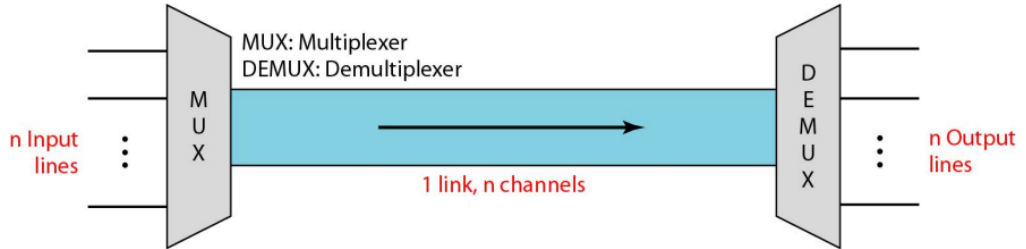


Multiplexing: Basic Terminology

- **Link:** the physical path – the actual medium (cable, fiber, air).
- **Channel:** a portion of a link that carries a transmission between a given pair of lines.
- One link can be divided into **many channels**, and many devices can share it at the same time.



Multiplexing-Demultiplexing



Categories of Multiplexing

- **Frequency-Division Multiplexing (FDM)** – analog technique, splits by frequency.
- **Wavelength-Division Multiplexing (WDM)** – FDM for light signals in fiber optics.
- **Time-Division Multiplexing (TDM)** – digital technique, splits by time.
 - Synchronous TDM
 - Statistical TDM

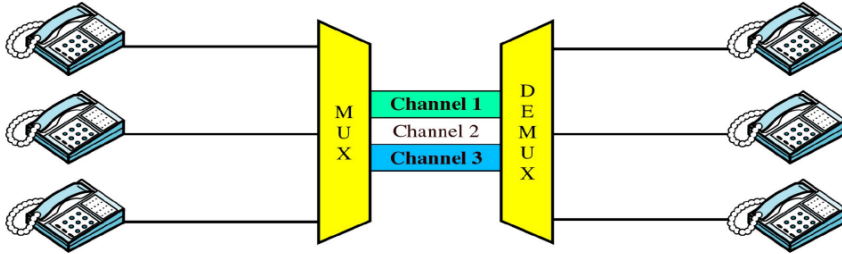


Frequency-Division Multiplexing (FDM)

- FDM is an **analog** multiplexing technique that combines analog signals.
- Each signal is modulated onto a **different carrier frequency**, so all signals travel over the link at the same time without overlapping.
- **Multiplexer (MUX)**: combines multiple transmission streams into a single stream (many-to-one).
- **Demultiplexer (DEMUX)**: separates the single stream back into its component transmissions (one-to-many) and routes them to the intended receivers.

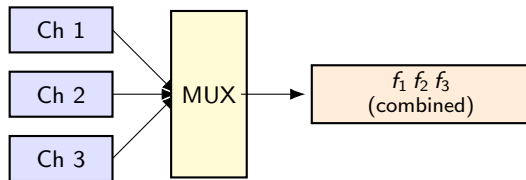


Frequency-Division Multiplexing (FDM)

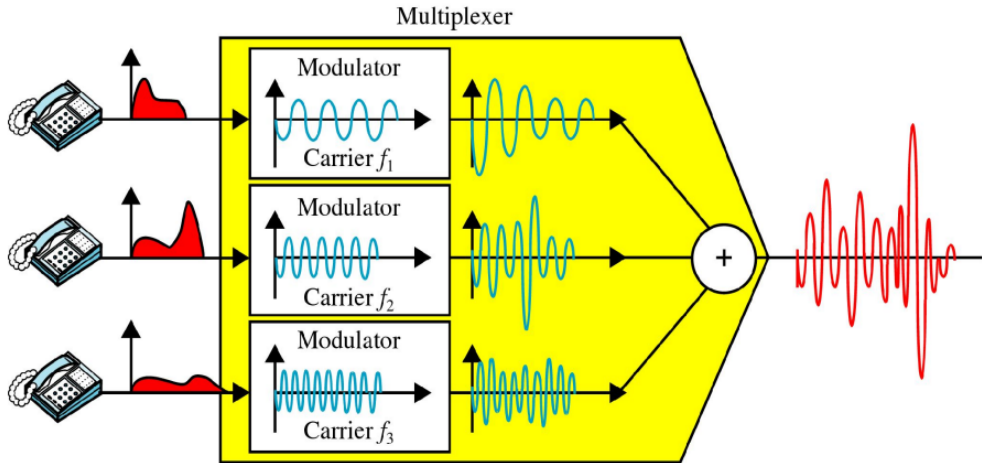


FDM: The Multiplexing Process

- Each input signal (e.g. each telephone call) generates a signal in a similar frequency range.
- These signals are shifted (modulated) onto different, non-overlapping carrier frequencies ($f_1, f_2, f_3, \dots$).
- The shifted signals are then combined into a single composite signal for transmission.



FDM: Multiplexing

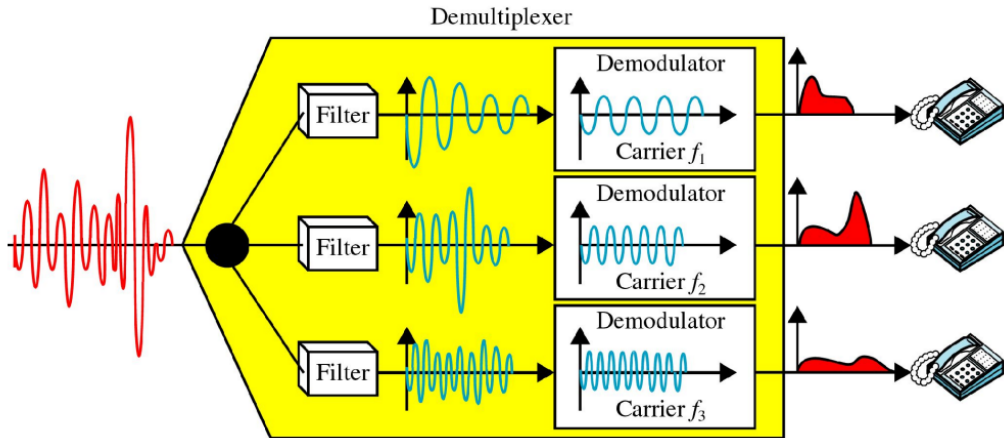


FDM: Demultiplexing

- The demultiplexer uses **filters** to decompose the composite signal into its constituent frequency bands.
- Each band is then demodulated to recover the original signal and passed to the waiting receiver.
- This is exactly the reverse of the multiplexing process shown on the previous slide.



FDM: Demultiplexing

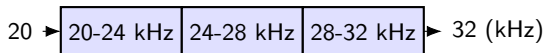


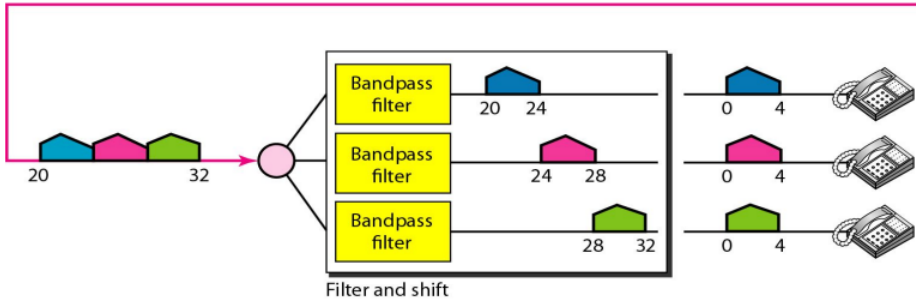
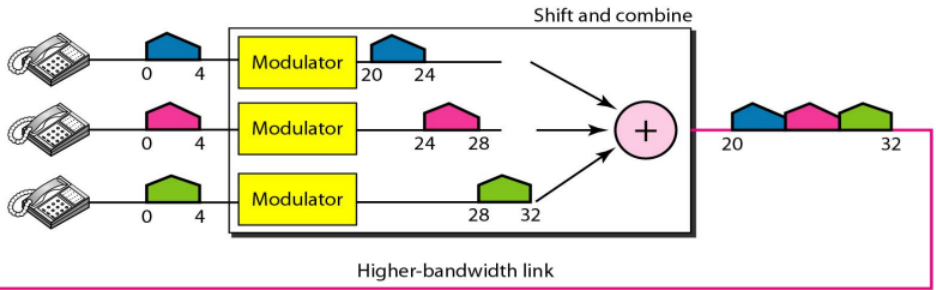
Example 6.1: FDM without Guard Bands

Problem: A voice channel occupies 4 kHz. Combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration (assume no guard bands).

Solution: We shift (modulate) each of the three voice channels to a different band:

- Channel 1 $\rightarrow$ 20–24 kHz
- Channel 2 $\rightarrow$ 24–28 kHz
- Channel 3 $\rightarrow$ 28–32 kHz





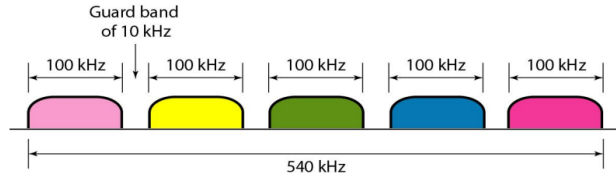
Example 6.2: FDM with Guard Bands

Problem: Five channels, each with 100 kHz bandwidth, are multiplexed together. What is the minimum bandwidth of the link if a 10 kHz guard band is needed between channels?

Solution:

- For 5 channels, we need at least **4 guard bands** (one between each adjacent pair).
- Required bandwidth:

$$5 \times 100 \text{ kHz} + 4 \times 10 \text{ kHz} = 540 \text{ kHz}$$



Note: guard bands waste bandwidth but prevent interference between channels.



FDM in Practice: Cable Television

- A coaxial cable has a bandwidth of approximately **500 MHz**.
- Each individual television channel requires about **6 MHz** of bandwidth.
- Theoretically, a single coaxial cable can carry:

$$\frac{500 \text{ MHz}}{6 \text{ MHz}} \approx 83 \text{ channels}$$

- This is a real-world, everyday example of FDM: many TV channels sharing one cable simultaneously, each on its own frequency band.

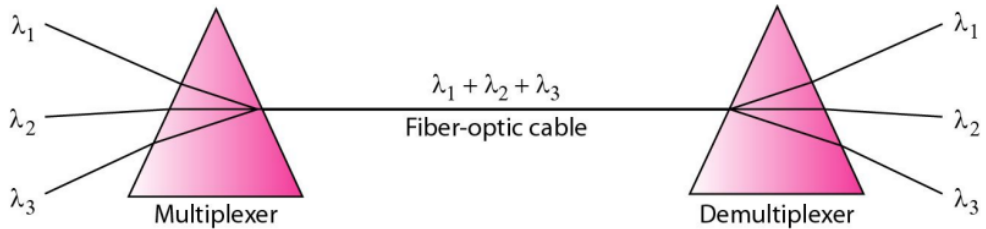


Wavelength-Division Multiplexing (WDM)

- WDM is conceptually **the same as FDM**, except that multiplexing and demultiplexing involve **light signals** transmitted through **fiber-optic** channels.
- Very narrow bands of light from different sources are combined into a wider band of light.
- Combining and splitting of light sources is handled by a **prism**, which bends a beam of light based on its angle of incidence and frequency.
- Key application: **SONET** (fiber-optic backbone networks).



Wavelength-Division Multiplexing (WDM)

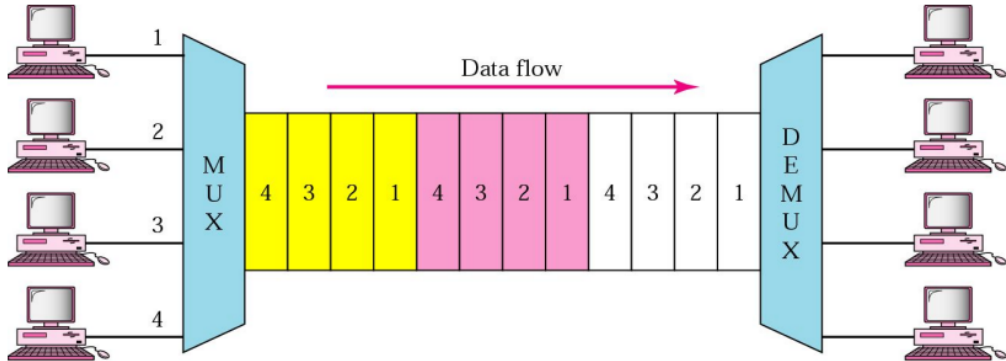


Time-Division Multiplexing (TDM)

- TDM is a **digital** process, used when the data-rate capacity of the transmission medium is **greater** than the data rate required by the sending/receiving devices.
- Several connections share the **high bandwidth of a link** by taking turns over **time** instead of frequency.
- Two schemes:
 - **Synchronous TDM**
 - **Statistical TDM**

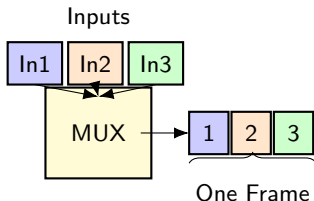


Synchronous TDM

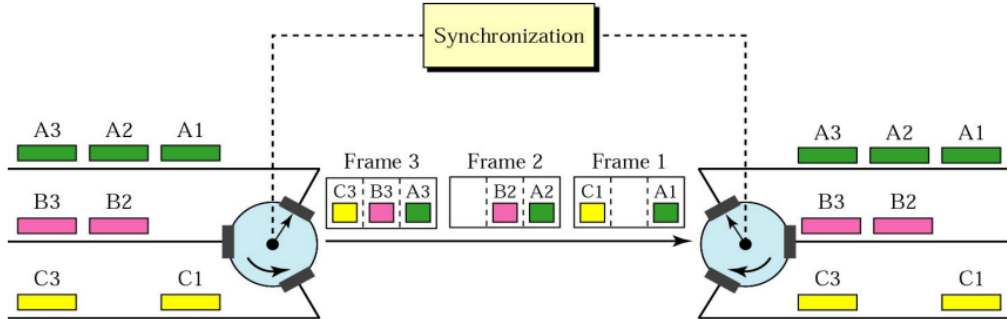


Synchronous TDM: Interleaving

- Synchronous TDM can be compared to a very fast rotating switch: one switch on the multiplexing side, one on the demultiplexing side, **synchronized**, rotating at the same speed but in opposite directions.
- As the switch opens in front of a connection, that connection gets the opportunity to send one unit onto the shared link – this process is called **interleaving**.
- If a source has **no data to send**, its slot in the output frame is left **empty** (wasted bandwidth).



Synchronous TDM: Interleaving



Example 6.3: Synchronous TDM Calculation

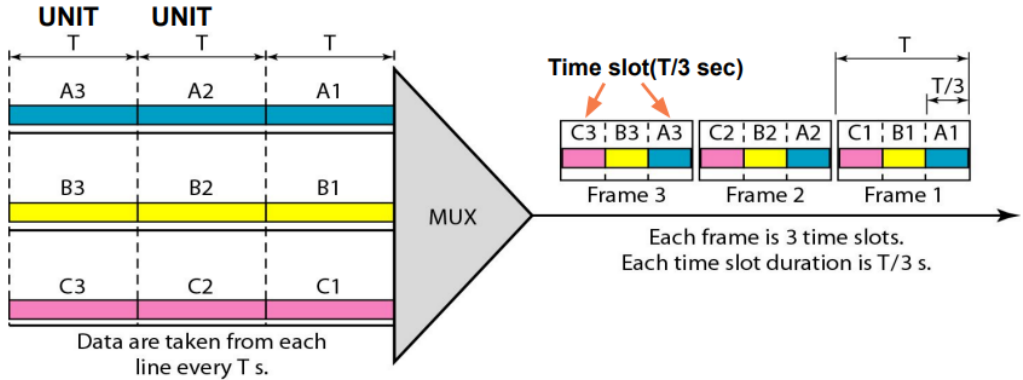
Problem: A multiplexer combines three channels. Each input connection has a data rate of 1-kbps. If 1 bit at a time is multiplexed, find:

- (a) duration of each input slot
- (b) duration of each output slot
- (c) duration of each frame

Solution:

	Quantity	Result
(a)	Input slot duration	$1/1\text{ kbps} = 1\text{ ms}$ (bit duration)
(b)	Output slot duration	$1\text{ ms}/3 = 1/3\text{ ms}$
(c)	Frame duration	$3 \times 1/3\text{ ms} = 1\text{ ms}$ (same as input unit)





Example 6.4: Synchronous TDM Calculation

Problem: Four 1-kbps connections are multiplexed together, one bit at a time. Find:

- (a) duration of 1 bit *before* multiplexing
- (b) transmission rate of the link
- (c) duration of a time slot
- (d) duration of a frame

Solution:

	Quantity	Result
(a)	Bit duration before mux	$1/1\text{ kbps} = 1\text{ ms}$
(b)	Link rate	$4 \times 1\text{ kbps} = 4\text{ kbps}$
(c)	Time-slot duration	$1\text{ ms}/4 = 250\text{ }\mu\text{s}$
(d)	Frame duration	$4 \times 250\text{ }\mu\text{s} = 1\text{ ms}$ (same as bit duration before mux)



Example 6.5: Synchronous TDM Calculation

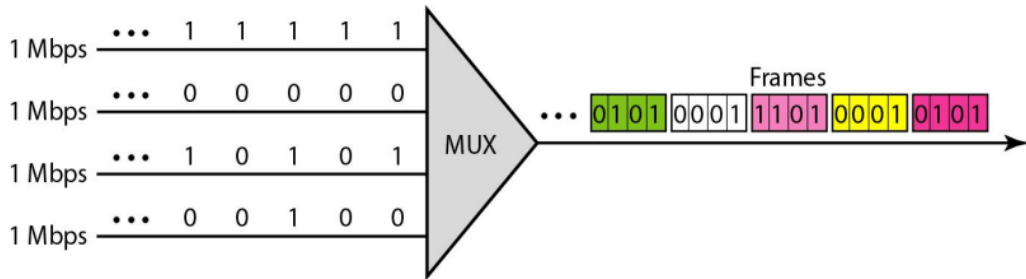
Problem: Four 1-Mbps connections are multiplexed synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find:

- (a) input bit duration
- (b) output bit duration
- (c) output bit rate
- (d) output frame rate

Solution:

Quantity		Result
(a)	Input bit duration	$1/1\text{Mbps} = 1\ \mu\text{s}$
(b)	Output bit duration	$1\mu\text{s}/4 = 1/4\ \mu\text{s}$
(c)	Output bit rate	$4 \times 1\text{Mbps} = 4\ \text{Mbps}$
(d)	Output frame rate	1,000,000 frames/s (same as any input rate)





Example 6.6: TDM with Byte-Level Multiplexing

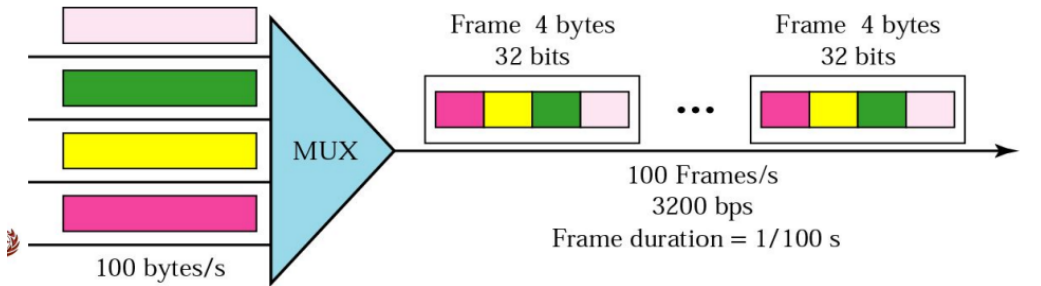
Problem: Four channels are multiplexed using TDM. Each channel sends 100 bytes/s and we multiplex 1 byte per channel. Show the frame traveling on the link, and find:

- (a) size of the frame
- (b) duration of a frame
- (c) frame rate
- (d) bit rate for the link

Solution:

Quantity		Result
(a)	Frame size	$4 \times 1 \text{ byte} = 4 \text{ bytes} = 32 \text{ bits}$
(b)	Frame duration	$1/100 \text{ bytes/s} = 10 \text{ ms}$ (same as 1 byte before mux)
(c)	Frame rate	100 frames/s (same as each input's byte rate)
(d)	Bit rate for the link	$100 \times 32 \text{ bits} = 3200 \text{ bps} = 3.2 \text{ kbps}$





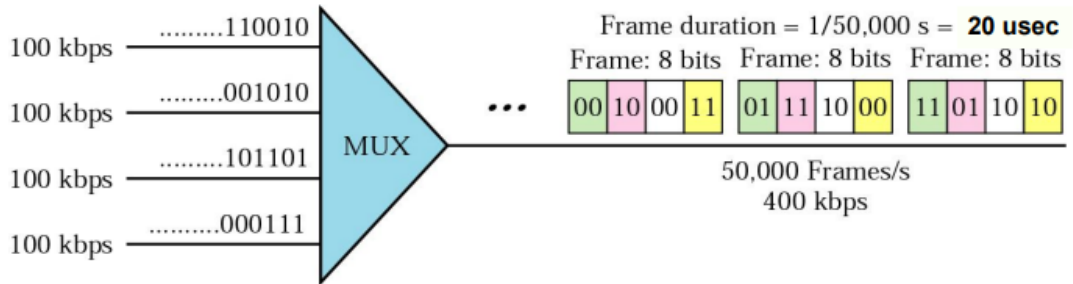
Example 6.7: TDM with 2-bit Time Slots

Problem: A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. Find the frame rate, frame duration, bit rate, and bit duration.

Solution:

	Quantity	Result
1	Frame rate	$100,000 \text{ b/s} / 2 \text{ bits} = 50,000 \text{ frames/s}$
2	Frame duration	$1/50,000\text{s} = 20 \mu\text{s}$
3	Bit rate	$50,000 \times 8 \text{ bits} = 400 \text{ kbps}$
4	Bit duration	$1/400,000\text{s} = 2.5 \mu\text{s}$





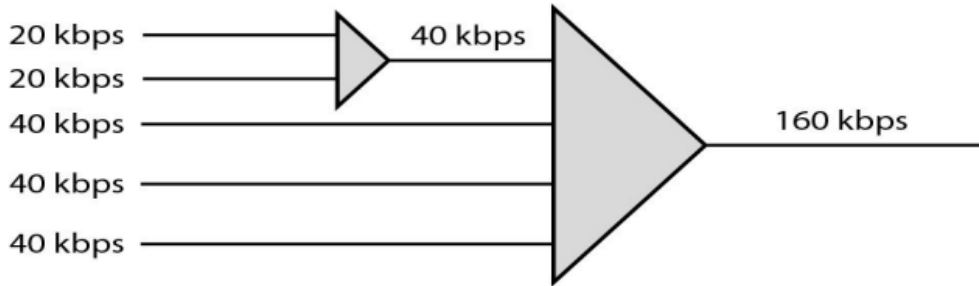
Synchronous TDM: Handling Unequal Data Rates

Real input lines rarely all have the **same** data rate. Three strategies handle this mismatch:

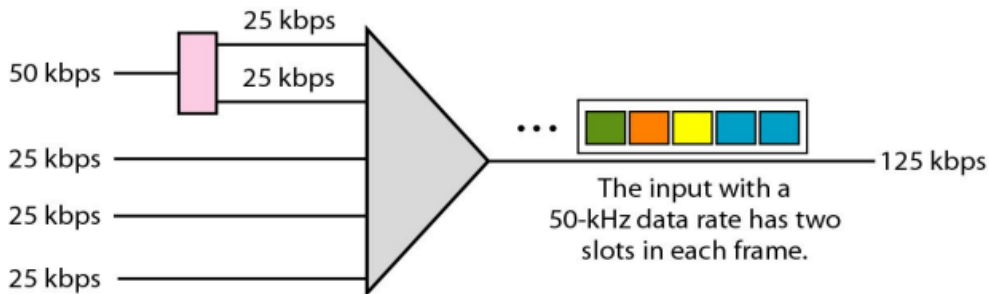
1. **Multilevel Multiplexing**: used when one input's rate is a **multiple** of another's – lower-rate lines are first combined together, then multiplexed again with the higher-rate lines.
2. **Multiple-Slot Allocation**: a faster input line is given **more than one slot** per frame (a serial-to-parallel converter splits it into multiple sub-streams first).
3. **Pulse Stuffing**: when rates are **not** clean multiples of each other, the highest rate is treated as dominant and **dummy (stuffing) bits** are added to the slower lines to match it.



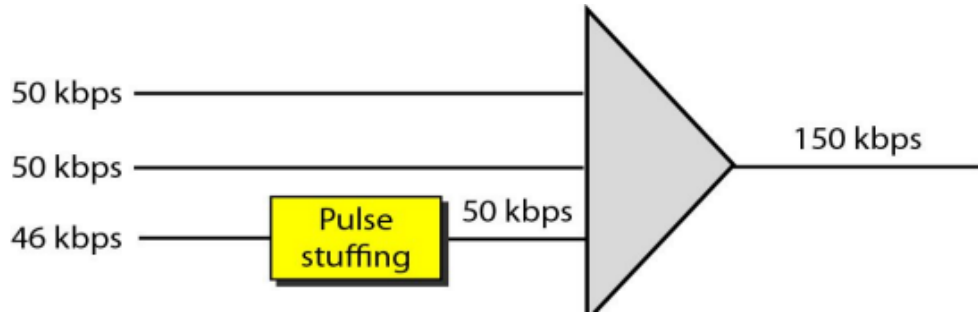
Handling Unequal Data Rates: Multilevel



Handling Unequal Data Rates: Multiple Slot



Handling Unequal Data Rates: Pulse Stuffing



Synchronous TDM: Framing Bits

- If the multiplexer and demultiplexer lose synchronization, a bit belonging to one channel could be delivered to the **wrong** channel.
- **Framing bits** let the demultiplexer stay synchronized with the incoming stream so it can identify time slots accurately.
- Typically, **1 synchronization bit** is added at the start of each frame, alternating between 0 and 1 (e.g. 1 0 1 0 1 0 ...).



Statistical TDM

- In **Synchronous** TDM, some slots go empty when a line has no data – this wastes bandwidth.
- In **Statistical** TDM, slots are allocated **dynamically**: no slot is left empty as long as **any** input line has data to send.
- **Addressing**: since slots are no longer fixed to a specific line, each slot must carry the **address** of its destination along with the data.
 - To address N possible output lines, we need $n = \log_2 N$ address bits.
- **No synchronization bit** is needed – since slot order isn't fixed, frames don't need to be synchronized the way they do in synchronous TDM.

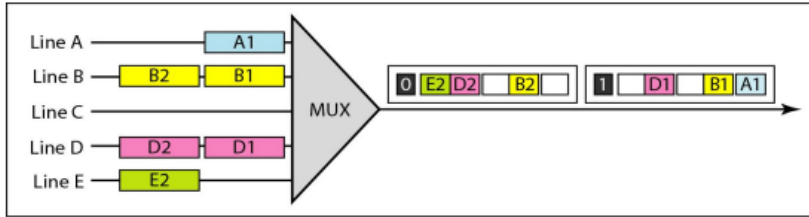


Synchronous vs. Statistical TDM

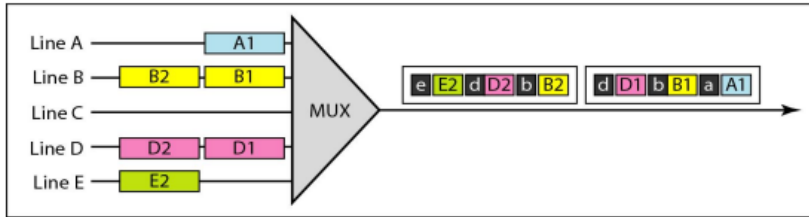
Synchronous TDM	Statistical TDM
Slots pre-assigned to each line Empty slots wasted if a line is idle	Slots allocated on demand No wasted slots (as long as some line has data)
No addressing needed (slot position = identity)	Address bits needed in each slot
Needs framing/sync bits	No framing/sync bits needed



Synchronous vs. Statistical TDM



a. Synchronous TDM



b. Statistical TDM

Practice Problems

- Theoretical:
Data Communication and Networking (4th Ed) – Behrouz A. Forouzan
Chapter 06 – 1, 3, 6
- Mathematical:
Data Communication and Networking (4th Ed) – Behrouz A. Forouzan
Chapter 06 – 9, 12, 15, 18



References

- Data Communication and Networking (4th Ed) – Behrouz A. Forouzan
Chapter 06 – 6.1 (Multiplexing)



Thank You

